



University of Al-Ameed
College of Medicine
Department of Family and Community Medicine



FUNDAMENTALS OF HUMAN NUTRITION AND DIETARY GUIDELINES

BY

NIBRAS TAREQ AL NAQEEB

CABMS/FM



Learning Objectives:

At the end of this lecture the student will be able to :

- Define nutrition and describe its relationship to health and well-being.
- List factors effecting on the nutritional status of individual
- Define some related terms to nutrition.
- Recognize the food pyramid and the purpose of its use.
- Demonstrate how to calculate the energy available from food

Definition:

Nutrition:

Defined as the science of food, the nutrients, and other substances, their action, interaction and balance in relation to health & disease, and the process by which they are used by the body.



Nutrition is essential for life, health, and development.

Proper nutrition is important throughout all life stages:

- Fetal growth.
- Birth and breastfeeding.
- Childhood and adolescence.
- Adulthood and old age.

A balanced diet is key to:

- Survival and physical growth
- Mental development
- Improved performance and productivity
- Overall health and well-being

Why Nutrition is Important?

1. Proper nutrition supports wellness:

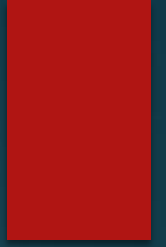
Wellness is more than the absence of disease

Physical, emotional, mental, psychological and spiritual health.

- Active process.

Critical components of wellness

- Nutrition
- Physical activity



2. Nutrition can prevent disease:

- Nutrient deficiency diseases:
scurvy, goiter, rickets
- Diseases influenced by nutrition:
chronic diseases such as heart disease
- Diseases in which nutrition plays a role:
osteoarthritis, osteoporosis

3. Obesity is a growing problem:

- Eating more calories than expended
- Risk factor for heart disease, stroke, type 2 diabetes, and some forms of cancer

4. Nutrition is an international goal to promote optimal health and disease prevention.



Healthy People 2030: Overarching Goals

- Promote **healthy, thriving lives and well-being** free from preventable disease, disability, injury, and premature death.
- **Eliminate health disparities** and improve health and health literacy for all.
- Create **social, physical, and economic environments** that support health and well-being.
- Promote **healthy development and healthy behaviors across all life stages**.

The Role of Nutrition in Health and Tissue Function

Nutrition is key for:

- **Building tissues:** Supplying the body with proteins, fats, and other materials necessary for creating and developing cells.
- **Repairing and renewing tissues:** Ensuring the body can regenerate and heal tissues through a consistent supply of nutrients.
- **Maintaining bodily functions:** Supporting organs and systems with essential vitamins, minerals, and energy to function properly

The Role of Nutrition in Health and Tissue Function

Health is influenced by:

- **Nutrient intake:** Ensuring a balanced diet that meets the body's requirements for essential nutrients like carbohydrates, proteins, fats, vitamins, and minerals.
- **Nutrient utilization:** The body's ability to absorb and effectively use nutrients for energy, growth, and repair.
- **Preventing deficiencies:** Avoiding health issues related to poor nutrition, such as weakened immunity, slowed growth, and chronic diseases.

What are the factors that affect the nutritional status of individuals?

- Age: Nutritional needs change with age.
- Gender: Men and women require different nutrients.
- Health Status: Consider medical conditions, allergies, or pregnancy.
- Activity Level: More active individuals need more calories and nutrients.
- Seasons: Adjust meals for seasonal food availability and energy needs.
- Cultural Habits: Incorporate traditional foods and eating patterns.
- Economic Situation: Plan meals that are nutritious yet budget-friendly.
- Food Availability: Ensure ingredients are accessible and affordable

Consequences On Health: Of Poor Nutrition

- ▶ 1. Cardiovascular disease (CVD)
- ▶ 2. Hypertension
- ▶ 3. Diabetes
- ▶ 4. Cancer
- ▶ 5. Osteoporosis
- ▶ 6. Problems of being overweight or obese
- ▶ 7. Mental Disorders

Terms Used In Nutrition

Nutrients: The nourishing chemical substances in food that provide energy and promote the growth and maintenance of human body.

The essential nutrients : Are those the body cannot make for itself in sufficient quantity to meet physiological needs, and which must therefore be obtained from food.

Criteria for an Essential Nutrient

A nutrient is considered **essential** when it meets these criteria:

1. **The body cannot synthesize it**, or cannot produce enough to meet its needs.
2. **It must be obtained from the diet.**
3. **Deficiency causes specific signs or impaired physiological function.**
4. **Providing the nutrient prevents or corrects the deficiency.**

Examples: Essential amino acids, essential fatty acids, vitamins, minerals, and water.

The Six Types of Nutrients

- Food nourishes the body with more than 45 different nutrients. These nutrients are grouped into six categories.

- ▶ Carbohydrates

- ▶ Proteins

- ▶ Fats

- ▶ Vitamins

- ▶ Minerals

- ▶ Water

- ▶ • Eating a variety of foods to provide these nutrients is essential to good health.

Nutrients can be divided into two main types:

1. Organic Nutrients: These contain carbon and are usually found in complex forms. They include:

Carbohydrates: Provide energy.

Fats: Supply long-term energy and help absorb vitamins.

Proteins: Build and repair tissues.

Vitamins: Support various body functions



2. **Inorganic Nutrients:** These do not contain carbon and are simpler in structure. They include:

Minerals: Essential for various bodily functions.

Water: Vital for hydration and numerous bodily processes

Classifications of nutrients

By function:

- ▶ Fuel Nutrients needed for energy: carbohydrates, fats, and proteins.
- ▶ Regulatory nutrients necessary to function normally with no caloric value: vitamins, minerals, and water

Classification of foods according to their functions:

PROTECTIVE FOODS

prevent diseases,
rich in vitamins and minerals.

include vegetables and fruits



BUILDING FOODS essential for growth and cell repair include various types of meat, poultry, fish, eggs, dairy products, and legumes such as beans, lentils, chickpeas, and kidney beans



ENERGY FOODS provide the body with the necessary energy for activity, movement, warmth, and essential bodily functions. These foods include various grains and their products, tubers such as potatoes and sweet potatoes, as well as sugars, animal fats, and various types of vegetable oils.



LEARN
EAT
WELL

Classifications of nutrients

2- By Amount:

Macronutrients needed in proportionally large

amounts daily: **carbohydrates, fats, proteins**, Water is also a macronutrient in the sense that you require a large amount of it, but unlike the other macronutrients, it does not yield energy. **se** that require a large amount of it, but unlike the other micronutrient

Micronutrients required in small amounts daily:

vitamins and minerals.

A. Macronutrients which include:

- Carbohydrates
- Protein
- Fat

B. Micronutrients which include:

1. Vitamins.
2. Minerals



EAT A VARIETY OF FOODS FROM THE FOOD GROUPS



Go!



Grow!



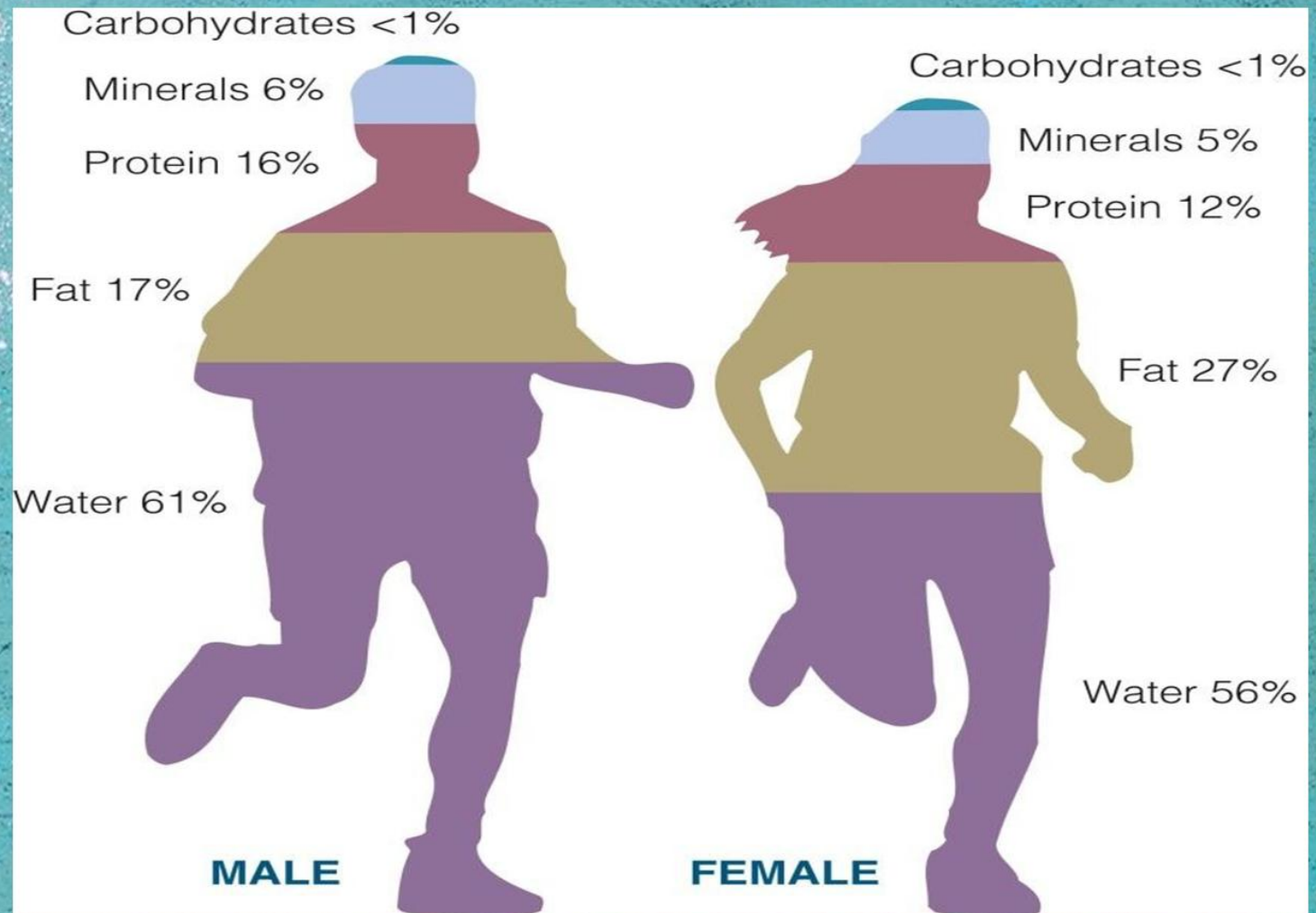
Glow!



water ;

- ▶ **World Health Organization (WHO)**
No single universal value (needs vary by climate, activity, age). Generally advises 2–3 liters/day for adults, adjusting for hot environments, exercise, pregnancy/lactation, or illness (e.g., fever, diarrhea).

Water



Characteristics of a Nutritious Food

- ▶ Adequate
- ▶ Balanced
- ▶ Moderate
- ▶ Varied

Estimated Average Requirement (EAR)

Average daily level of intake estimated to meet the requirements of 50% of healthy individuals; in a particular life stage and gender group

Adequate Intake (AI):

An observational standard that is used when insufficient data is available to determine an RDA from an existing EAR. Because it is set using presumable healthy groups of individuals, the AI is expected to meet or exceed the actual nutrient requirement in all healthy members of a specific life stage and gender group.

Enough scientific evidence → establish EAR → calculate RDA

Not enough evidence to establish EAR → use AI instead

Recommended Dietary Allowance (RDA)

- Daily amount of a nutrient considered adequate to meet known nutrient needs of almost 98% of all healthy people .

Nutrient allowances are categorized into (17) classifications based on age & sex. RDA does not provide the needs that have been altered as a result of disease states, chronic usage of certain drugs, or other factors that require specific individual attention.

	Age (Years)	Weight (kg)	Ca (mg)	Fe (mg)	Mg (mg)	Zn (mg)
Infants	0- ¹ / ₂	6	360	10	60	3
	¹ / ₂ -1	9	540	15	70	5
Children	1-3	13	800	15	150	10
	4-6	20	800	10	200	10
	7-10	30	800	10	250	10
Males	11-14	44	1200	18	350	15
	15-18	61	1200	18	400	15
	19+	67+	800	10	350	15
Females	11-18	44-54	1200	18	300	15
	19+	58	800	18(10)*	300	15
Pregnant			1200	18+**	450	20
Lactating			1200	15	450	25

Recommended (Daily) Dietary Allowances (RDA) of The Food and Nutrition Board
(Published by the National Academy of Science, Washington, DC, U.S.A).



Nutrient intake below 70% of RDA are frequently considered to be the least RDA levels of nutrients below which put an individual at risk of clinical deficiency

Food pyramid (nutrition)

- The food pyramid is a triangular guide that categorizes foods based on their health benefits. As you move up, foods become less healthy and should be consumed in moderation. It provides guidelines on **portion sizes, balancing food types**, and **avoiding fatty foods** to ensure nutrient intake, maintain health, and prevent obesity. The first pyramid was published in Sweden in 1974

▶ **USDA Food Guide Pyramid (1992)**

- First U.S. pyramid (horizontal layers): base **6–11 grain servings**, mid-tiers fruits/vegetables, then dairy & meat groups; top fats/sweets “use sparingly.”

▶ **MyPyramid (2005):** what changed? Vertical colored bands (proportionality), figure on steps (activity), less intuitive for the public without food images; still limited on food quality guidance.

▶ **MyPlate (2011): from pyramid to plate**

- Replaced pyramid with a **plate divided into fruits, vegetables, grains, protein + dairy “cup.”**

fats, oils, and sweets,
use sparingly

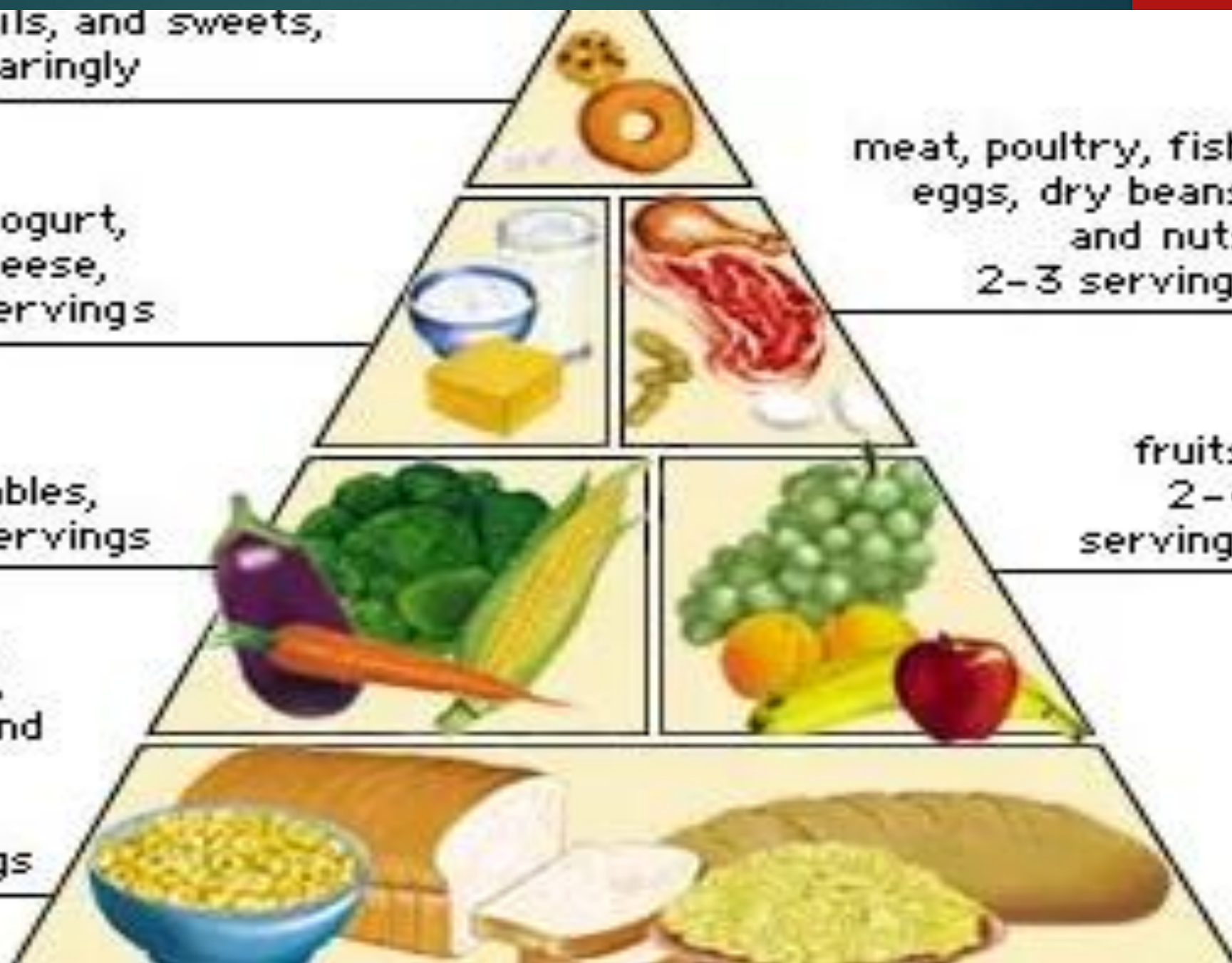
milk, yogurt,
and cheese,
2-3 servings

vegetables,
3-5 servings

bread,
cereal,
rice, and
pasta,
6-11
servings

meat, poultry, fish,
eggs, dry beans,
and nuts,
2-3 servings

fruits,
2-4
servings



MyPyramid

STEPS TO A HEALTHIER YOU

MyPyramid.gov



GRAINS

VEGETABLES

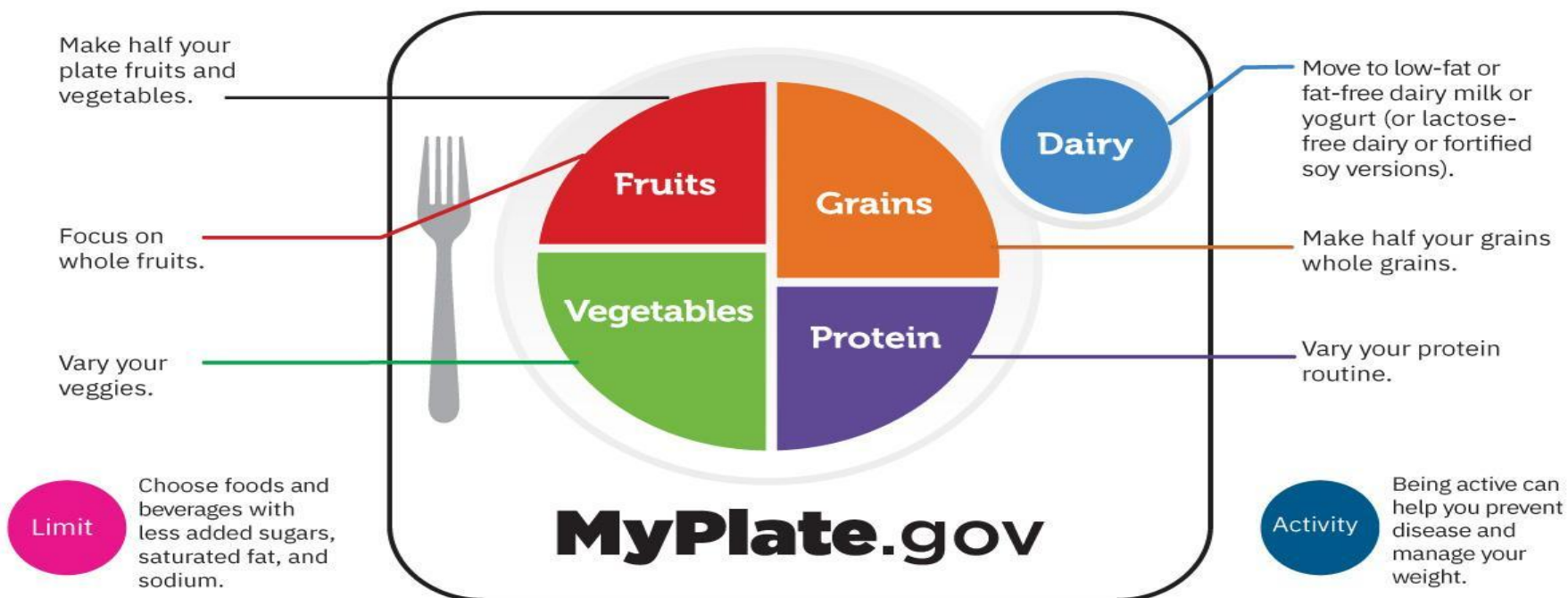
FRUITS

MILK

MEAT & BEANS

Start *simple* with **MyPlate**

Healthy eating is important at every life stage,
with benefits that add up over time, bite by bite. Small changes matter.

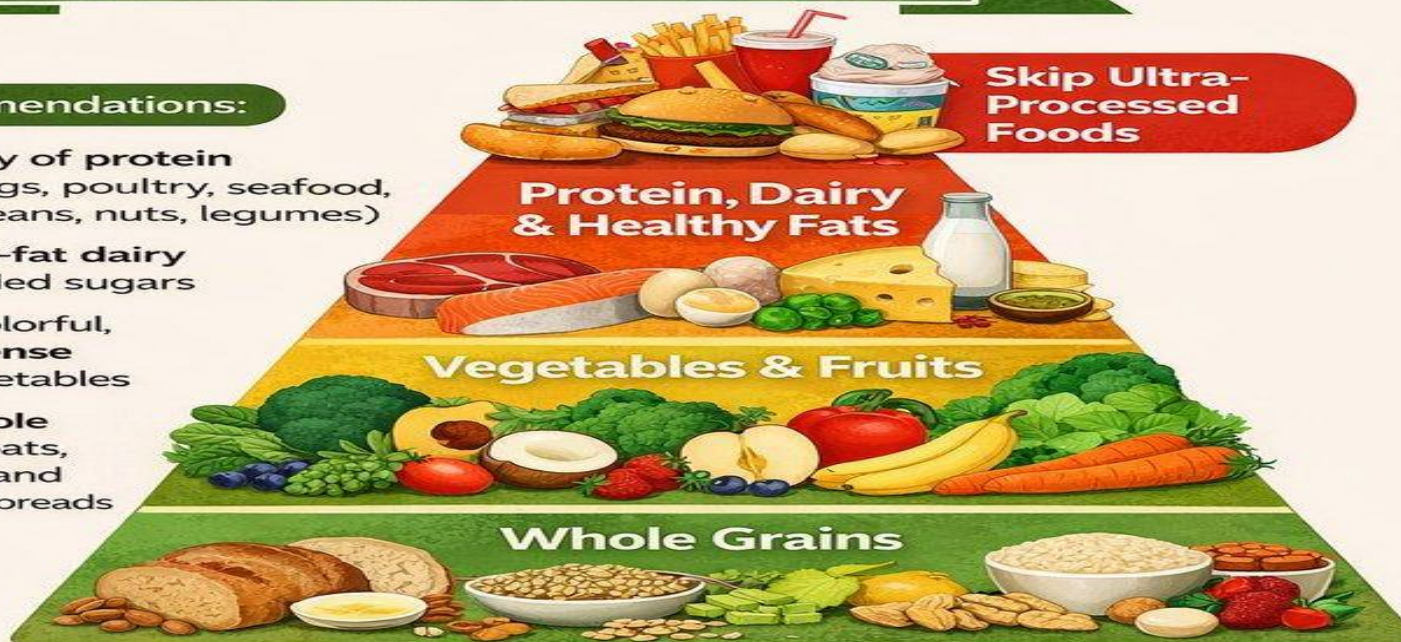


New 2025-2030 Government Dietary Guidelines

Eat More Real Food

Key Recommendations:

- ✓ Eat a variety of **protein** sources (eggs, poultry, seafood, red meat, beans, nuts, legumes)
- ✓ Choose **full-fat dairy** without added sugars
- ✓ Consume colorful, **nutrient-dense** fruits & vegetables
- ✓ Choose **whole grains** like oats, brown rice, and whole-grain breads



Prioritize nutrient-dense, whole foods and switch to balanced, less processed meals. Reduce refined grains, added sugars, and unhealthy oils and additives.

Learn more at RealFood.gov

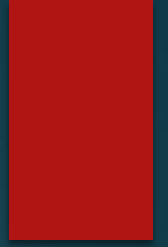


Dietary Guidelines for Americans 2025-2030,
realfood.gov



Prioritize nutrient-dense, whole foods and switch to **balanced, less processed** meals. Reduce refined grains, added sugars, and unhealthy oils and additives.

Food pyramid:



Recommendation of what to eat based on dietary Guidelines

Food pyramid from top to bottom :

- • Fats
- • Dairy
- • Meat
- • Fruits & Vegetables
- • Grains

Serving size

A serving size: is the suggested amount of food or beverage to consume at a meal or in a snack

Serving sizes help to regulate the amount of calories consumed each day; they also encourage eating a wide variety of foods to completely meet nutritional needs.

The food from **MILK GROUP**, are a major source of calcium, Protein & riboflavin. [2-3servings/ day]
Serving sizes include 1 cup milk, 1 cup yogurt.
Take some dice to get an idea of the serving size for cheese:
(1 oz=28 g). of cheese is the size of four dice



Meat Group[2-3servings/ day] ►

Items in the meat group supply protein, fat, iron and other minerals

Red meat includes beef, lamb, venison , (dry beans, eggs& nuts group). [2-3 servings/ in day]

The recommended daily intake of meat or other protein equivalents ranges from 6 to 9 oz. a day based on gender, age and calorie needs.

Three ounces(serving size) of meat is approximately the size of the palm of hand or a deck of card

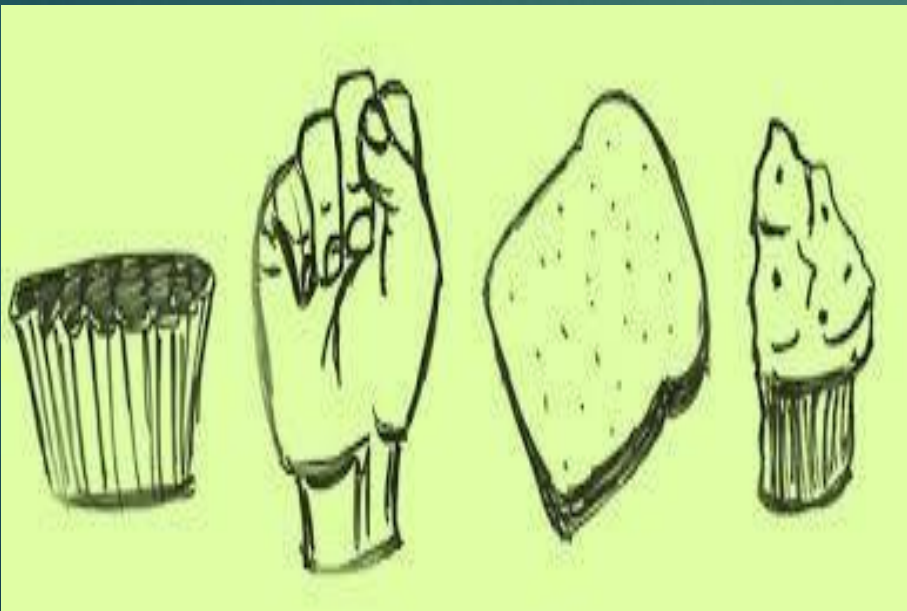


Bread and cereal group

Provide carbohydrate, several B vitamins & iron.

{6-11 servings/ day}

Some examples of serving sizes include, **1/2 cup** cooked rice which is about the size of a **tennis ball**, a slice of bread the size of **an audio-cassette tape**, and a cup of pasta/spaghetti or cereal, both about the size of a **fist**.





Fruits & Vegetables

Are rich in vitamin C & precursors of vitamin A.

Vegetables group[3-5 servings/ day]

Fruit group [2-4 servings/ day]. Eating a diet rich in fruits, vegetables and whole grains is important to maintaining good health.

A tennis ball is a good visual for one medium piece of fruit.

A serving of cooked vegetables is $\frac{1}{2}$ cup while a serving of raw leafy vegetables is 1 cup





HAND:

Breads



PALM:

Meats



FIST:

**Veggies, Rice,
Pasta, Fruits**



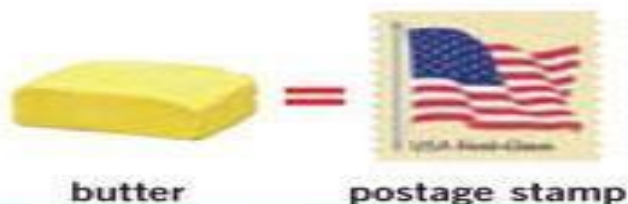
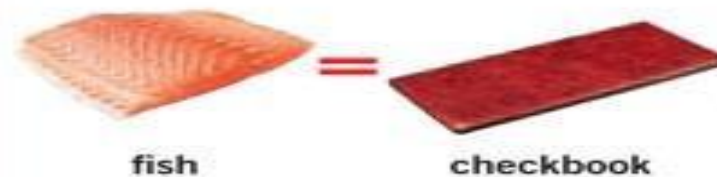
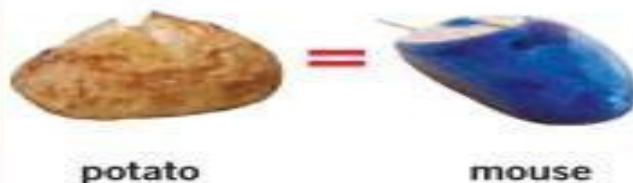
FINGERTIP:

Fats (butter)

© Grand-Slam-Weight-Loss-Tips.com 2012

SIZE IT RIGHT

A guide (based on standards that most nutritionists follow) to what one serving should look like.



HOW TO CALCULATE THE ENERGY AVAILABLE FROM FOODS

To calculate the energy available from a food, multiply the number of grams of

Carbohydrate 4 kcal/g,

Protein 4 kcal/g

Fat 9 kcal/g.

Then add the results together

For example,

1 slice of bread with 1 tablespoon of peanut butter on it contains 16grams carbohydrate, 7 grams protein, and 9 grams fat:

16g carbohydrate X 4 kcal/g = 64kcal

7g protein X 4kcal/g = 28kcal

9g fat X 9kcal/g = 81kcal

Total = 173kcal



To determine the percentage of calories from fat.

For Example, divide the 81fat kcalories by the total 173 kcalories:

$81 \text{ fat kcal} / 173 \text{ total kcal} = 0.468$ (rounded to 0.47)

Then multiply by 100 to get the percentage:

$0.47 \times 100 = 47\%$



Homework;

Calculate the energy available from a bean burrito with cheese (55 grams carbohydrate, 15 grams protein, and 12 grams fat).

Determine the percentage of kcalories from each of the energy nutrients

Any Question

